

The Lumen Medical Library: Digital Library Plan

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Introduction

This Digital Library Plan provides a blueprint for a fictitious medical library called the Lumen Medical Library. As a university science library, the Lumen Medical Library serves medical and nursing students. As a result of this very specific population, the sorts of materials provided by the library need to be specialized, equipping students with peer-reviewed medical databases and textbooks in the form of e-Books. In order to provide access and usability of this curated digital selection to students, the library will use a Digital Access Management system will be chosen in tandem with a metadata schema to ensure interoperability. Finally, user interface sketches will be provided that demonstrate preliminary designs of homepages for both mobile and desktop environments.

Collection in Repository

The Lumen Medical Library hosts two collections that are centralized into digitized repositories. The first collection includes hundreds of rare books printed before 1800 and the second includes archives of letters, case books, and student notebooks that document the professional practice of doctors. Digitized collections of historic importance offer invaluable opportunities for students to learn. Librarians working with the collection must be able to verbalize the important roles that digital collections have educationally. Medicine has made large strides in previous years, not always in the correct direction, and historic collections help us rationalize this (Thornton, 1987). John L Thornton writes about this, saying, “To confirm that one is really breaking fresh ground one must endure that the work of previous research workers, and particularly of the pioneers, has been fully appreciated. Did they make erroneous assumptions which have led their successors astray? Did their writings contain

clues to lines of research which could only be identified as our knowledge expanded? These are examples of questions which only can be solved by historical research.”

Digital Journals

Various collections of digital journals have also been selected for Lumen Medical Library. The selection of digital journals was heavily scrutinized in relation to the ALA Academic Library Selection Criteria guidelines which include bullet-points such as currency and timeliness of material, accuracy and dept of material, and coverage appropriate to the level of study of a subject (*Selection Criteria | ALA*, n.d.). Digital journal databases are divided into two categories: paid journals and free, open-access journals. Selecting from both of these categories is advantageous for a digital library as open-access journals bring down costs for the library acquisitions and paid journals provide a depth of literature often not accessible for open-access journals. Paid journals include:

- Ovid: provides access to a range of databases with a powerful search engine (Brunskill, n.d.)
- AccessMedicine: provides medical information, eBooks, drug monographs, videos, and images (Cassidy, n.d.)
- DynaMed: primary case database (Barton, 2005)
- Public Health Database: provides public health literature, interdisciplinary (*Public Health Database*, n.d.)
- TRIP Pro Database: health science database with that grades the quality of evidence and links to the highest level of evidence available (“What Is the Difference between Trip and PubMed?,” 2023)
- Web of Science: indexed older publications, index and archived to 1900 (Falagas et al., n.d.)

Free, open-access journals include:

- PubMed: provided by the US National Institute of Health, full-text archive (*PubMed Central*, n.d.)
- CINAHL: focus on nursing and allied health disciplines, includes computer programs (Porben, n.d.)
- Embase: includes a larger selection of international journals (Tuttle, n.d.)
- MedLine Plus: focus on information for patients (*MedlinePlus*, n.d.)

The combination of paid and free open-access journals ensures that the library's budget is taken into consideration, and that student have access to a plethora of pertinent information to their studies as misinformation can large impacts.

E-Books

A variety of e-Books have also been selected for the Lumen Medical Library's digital book collection to aid students in their academic endeavors. E-books are advantageous for a digital collection as they minimize the costs associated with maintaining physical collections (Novak et al., n.d.). Criteria for book selection include: relevancy, ability to be used in a variety of classes, and expertise in content.

E-books selected include:

- Carol Boswell and Cannon, Sharon, 2022: Introduction to Nursing Research: Incorporating Evidence-Based Practice with Navigate Advantage Access
- David Celentano and Moyses Szklo, 2018: Gordis Epidemiology
- Dustin T. Duncan, et al., 2024: The Social Epidemiology of the COVID-19 Pandemic

- Heather R. Hall and Roussel, Linda A., 2020: Evidence-Based Practice: An Integrative Approach to Research, Administration, and Practice
- Richard J. Hayes and Moulton, Lawrence H., 2022: Cluster Randomized Trials (Chapman & Hall/CRC Bio-statistics Series)
- Kathryn H. Jacobsen, 2020: Introduction to Health Research Methods: A Practical Guide
- Linda Young Landesman, 2021: Landesman's Public Health Management of Disasters: The Practical Guide
- Margaret B. Nolan, et al., 2023: Chronic Disease Epidemiology, Prevention, and Control
- Lisa M. Sullivan, 2023: Essentials of Bio-statistics in Public Health
- Wolfe, et al., 2022: Interdisciplinary Pediatric Palliative Care

A wide range of e-Books are critical to be relevant to the highest number of students possible.

System & Metadata Schema

In order to showcase all of the content carefully chosen in the above sections, the Lumen Medical Library has chosen CONTENTdm as its digital asset management system. CONTENTdm is an OCLC product that is quite powerful, with the ability to import metadata and items, manage collections, and configure websites that allow for dynamic website configurations (Martinson, n.d.). OCLC maintains servers and software stored on CONTENTdm while also backing up the data. By having all of this data maintained on a cloud server, those with access are able to access collections via the internet, needing only a web browser to access digital collections. Staff will have specific designations that give them varying administrative control that is decided by those with full

administrative control (*Using CONTENTdm | CARLI*, n.d.). CONTENTdm offers multiple options for harvesting metadata, these features ensure that enhances discoverability, interoperability, and streamlines data management. One feature is use of a tool called the *Digital Collection Gateway*, which allows institutions to synchronize their metadata to WorldCat. CONTENTdm collections can also be configured to be used as a repository for Open Archives Initiative Protocol for Metadata Harvesting (OAI-PMH) (*Harvesting*, 2018).

Another important consideration is choice in metadata schema in order to handle items in the collection. Dublin Core has been chosen as it offers ease of use, wide applicability, and syntax independence. As a medical research library, the Lumen Medical Library, a major advantage of Dublin Core is the ability to use Darwin Core as an extension. Darwin Core (DwC) was create to facilitate the documentation of representations of biodiversity and can be applied to medical data in order to document taxonomy, occurrence, events, and locations. This is an excellent application in relation to environmental human health, disease outbreak tracking, and public health research (Brenskelle et al., 2022). Pivoting back to Dublin Core, as a metadata schema it is often criticized for ambiguity and complex descriptions; however, in combintion with Darwin Core, the grand majority of items in the Lumen Medical Library collection would have their metadata documented adequately in versions that allow metadata to be harvested appropriately (Kiramang, 2008).

Sketches

Featured in this section are preliminary sketches of the Lumen Medical Library webpage interfaces for both desktop and mobile formats. For both versions of the webpage interfaces multiple

features such as centralized search functions and ability to find collections, e-Books, and digital repositories easily.

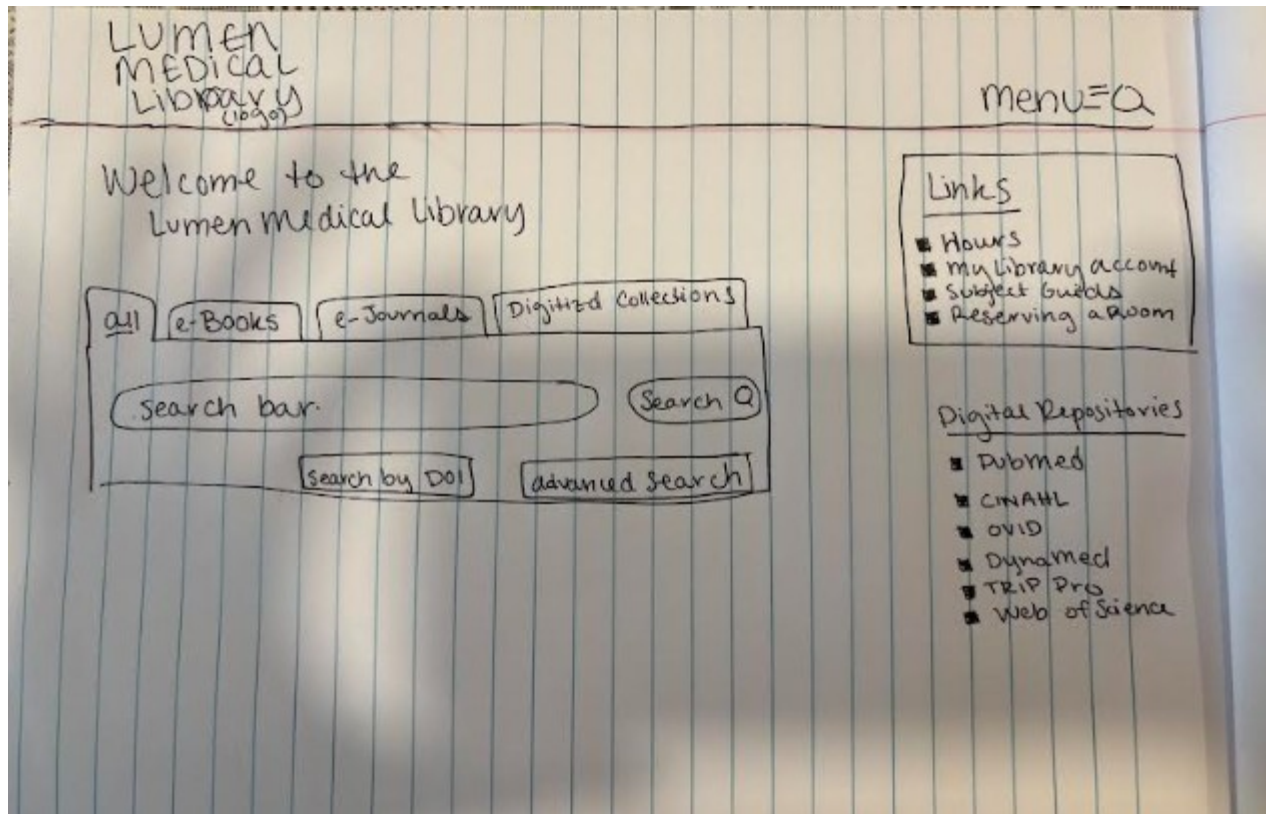


Figure 1: Lumen Medical Library Desktop Webpage Preliminary Sketch

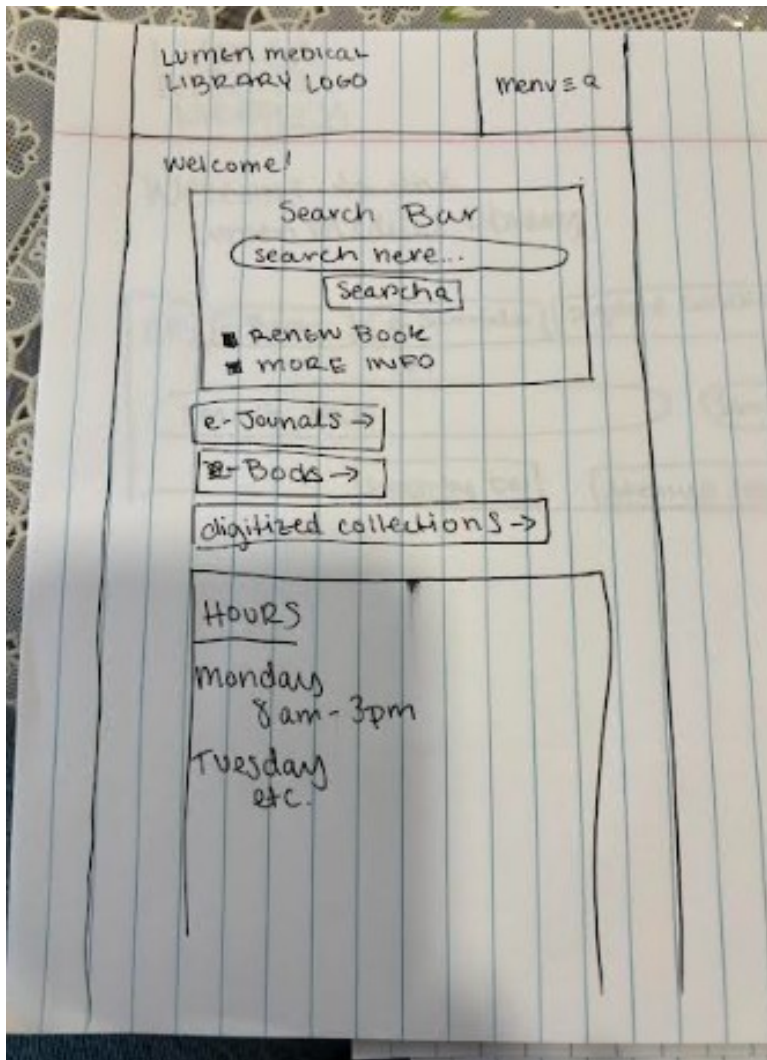


Figure 2: Lumen Medical Library Mobile Webpage Preliminary Sketch

Bibliography

- Barton, H. (2005). DynaMed. *Journal of the Medical Library Association*, 93(3), 412–414.
- Brenskelle, L., Wieczorek, J., Davis, E., Wallis, N. J., Emery, K., LeFebvre, M. J., & Guralnick, R. (2022). A community-developed extension to Darwin Core for reporting the chronometric age of specimens. *PLoS ONE*, 17(9), e0261044. <https://doi.org/10.1371/journal.pone.0261044>
- Brunskill, K. (n.d.). *Guides and databases: Ovid databases: What is Ovid?* Retrieved March 13, 2025, from <https://library-guides.ucl.ac.uk/OvidSP/what-is-OvidSP>
- Cassidy, J. (n.d.). *Pumerantz Library Research Guides: Database Information: AccessMedicine*. Retrieved March 13, 2025, from https://westernu.libguides.com/database_information/accessmedicine
- Falagas, M. E., Pitsouni, E. I., Malietzis, G. A., & Pappas. (n.d.). *Comparison of PubMed, Scopus, Web of Science, and Google Scholar: Strengths and weaknesses*. <https://doi.org/10.1096/fj.07-9492LSF>
- Harvesting*. (2018, February 2). OCLC Support. https://help.oclc.org/Metadata_Services/CONTENTdm/CONTENTdm_Administration/Server_Administration/020Harvesting
- Kiramang, K. (2008). *Comparative Analysis on Dublin Core (DC) and Metadata Object Description Schema (MODS)*.
- Martinson, T. (n.d.). *LibGuides: Digital Initiatives: CONTENTdm*. Retrieved March 14, 2025, from <https://library-nd.libguides.com/digitalinitiatives/contentdm>
- MedlinePlus: About MedlinePlus*. (n.d.). Retrieved March 13, 2025, from <https://medlineplus.gov/about/>

Novak, J., Ohler, A. L., & Day, A. (n.d.). Ebook Collection Development in Academic Libraries:

Examining Preference, Management, and Purchasing Patters. *Choice White Paper*.

https://www.choice360.org/wp-content/uploads/2020/09/Choice-White-Paper-Incorporating-Ebooks-into-Collection-Development_LO-RES_092820.pdf

Porben, E. (n.d.). *LibGuides: CINAHL Guide: Why Use CINAHL?* Retrieved March 13, 2025, from

<https://eguides.barry.edu/c.php?g=939791&p=6775361>

Public Health Database. (n.d.). Retrieved March 13, 2025, from

<https://about.proquest.com/en/products-services/publichealth/>

PubMed Central: About PMC. (n.d.). PubMed Central (PMC). Retrieved March 13, 2025, from

<https://pmc.ncbi.nlm.nih.gov/about/intro/>

Selection Criteria | ALA. (n.d.). Retrieved March 13, 2025, from

<https://www.ala.org/tools/challengesupport/selectionpolicytoolkit/criteria>

Thornton, J. L. (1987). The importance of the study of the history of medicine. *Health Libraries*

Review, 4, 139–140.

Tuttle, B. (n.d.). *LibGuides: Database Search Tips: Embase*. Retrieved March 13, 2025, from

<https://guides.mclibrary.duke.edu/searchtips/embase>

Using CONTENTdm | CARLI. (n.d.). Retrieved March 14, 2025, from

<https://www.carli.illinois.edu/products-services/contentdm/using-contentdm>

What is the difference between Trip and PubMed? What ChatGPT thinks. (2023, September 7). *Trip*

Database Blog. <https://blog.tripdatabase.com/2023/09/07/what-is-the-difference-between-trip-and-pubmed-what-chatgpt-thinks/>